

SHAMIL K V

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Education

BS-MS in Mathematics (Minor in Economic Sciences)

August 2019 – June 2025

Indian Institute of Science Education and Research (IISER), Bhopal

CGPA: 7.55

Relevant Courses: Econometrics, Probability Theory and Markov Processes, Numerical Analysis, Microeconomics, Macroeconomics, Applied Production Analysis

Diploma in Data Science

April 2025

Indian Institute of Technology (IIT) Madras

CGPA: 8.09

Certifications

NISM Series VIII: Equity Derivatives Certificate

Aug 2025

National Institute of Securities Markets (NISM)

Score: 85.25%

Experience

Pashupati Capital Services Private Limited, Mumbai (Goregaon)

Aug 2025 – Present

Quantitative Researcher

- Performed statistical and time-series analysis on **mid-frequency NSE equities, options, and futures data**, including **Nifty and Sensex derivatives**, to study volatility dynamics, liquidity, and regime behaviour.
- Designed a **quantitative backtesting framework** for **options strategies** (straddles, strangles, iron condors, butterflies, and spreads), evaluating **standalone and portfolio-level performance** across **volatility regimes** using **risk-adjusted metrics** (Sharpe ratio, Sortino ratio, maximum drawdown, and win rate).
- Implemented an automated **paper trading and execution-testing framework** using the **XTS API** to assess execution feasibility and calibrate slippage and fill assumptions.
- Designed and deployed **2 live trading strategies** driven by **volatility and open-interest signals**, building an end-to-end research and execution pipeline and extending the framework to additional signal classes.
- Designed a **scalable research and trading infrastructure** by integrating a **Kafka**-based streaming pipeline with **InfluxDB** for **real-time market data ingestion** and paper trading, alongside a **PostgreSQL**-based framework for **historical data management and backtesting**, enabling **multi-timeframe analysis** and efficient strategy development.
- Developed a **microstructure**-driven framework to capture **bid-ask spread** profits, leveraging **VWAP** and **relative spread** to identify trading opportunities.

Technical Skills

Programming: Python, SQL, MATLAB, R, C, LaTeX

Databases: PostgreSQL, InfluxDB (time-series)

Tools & Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Excel, STATA

Finance: Black-Scholes, Binomial Trees, Duration, Convexity, Yield Curves, VaR, CVaR

Quant Techniques: Stochastic Differential Equations, Ito's Lemma, Monte Carlo Simulation, Time-Series Modeling, Backtesting of Systematic Strategies

Modeling: EDA, Supervised Learning, Reinforcement Learning, Feature Engineering

Quantitative Projects

Stochastic Optimal Control in Finance (Master's Thesis)

2024–2025

Supervisor: Dr. Somnath Pradhan

- Established **weak convergence** of **discrete-time Markov chain approximations** to **continuous-time stochastic control problems** in finite horizon settings.
- Applied **dynamic programming** and the **Hamilton–Jacobi–Bellman (HJB) equation** to solve **Merton's optimal investment–consumption problem** and **time-consistent mean–variance portfolio optimization** within a stochastic control framework.
- Implemented **Q-learning** and **policy iteration** algorithms using NIFTY 50 and fixed deposit data for empirical validation.

Portfolio Optimization: Mean–Variance & Semi–Variance (L1-Penalized)

2026

- Built a robust **mean–variance optimization framework** with **L1-penalized downside risk control**, preserving **convexity**, and enabling **MINLP-based semi–variance formulations** for **asymmetric risk modeling**.
- Developed **multi-asset portfolios** (NIFTY 500 equities, gold, silver, and index ETFs) and conducted performance analysis using **Sharpe ratio, Information ratio, and maximum drawdown**.
- Implemented a **weekly rolling-window rebalancing strategy**, achieving a **7.6% annualized return**, while improving **portfolio efficiency** using **factor-based modeling** and **robust covariance estimation techniques**.

Machine Learning Projects

Bank Telemarketing Response Modeling

2024

- Developed and optimized classification models (**Logistic Regression, Random Forest, XGBoost**) to improve client targeting and lead conversion.
- Achieved **75% accuracy** and assessed model performance using **ROC-AUC, Precision, Recall, and F1-score**.